

MARIO ZAMUDIO

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Professional Summary

Solution Architect and ML/Data Engineer with 15 years of hands-on delivery for Bell Canada, AT&T, and Claro — now architecting cloud-native integration and AI systems at Corus Consulting (Sep 2025, remote). Completed a Master of Data Analytics at the University of Niagara Falls Canada (GPA 4.13, President's Distinction List all terms), with a Capstone deploying a production-grade ML system: Random Forest (AUC 0.98), K-Means segmentation, and a live Streamlit app on Google BigQuery. Expert in enterprise integration (IBM webMethods on Kubernetes/Helm, TIBCO, OpenShift), Big Data ETL (Hadoop, Spark, Hive), and AI-native engineering (Claude Code, LLM agents, GitHub Actions CI/CD). Available for mid-senior Solution Architect, Data Engineer, and ML Engineer roles across Canada.

Technical Skills

- **Languages & ML:** Python, SQL, R, scikit-learn, Pandas, NumPy, statsmodels, Java, Shell, C++
- **Data & BI:** Google BigQuery, Power BI (DAX), Tableau, Streamlit, Chart.js, MicroStrategy, Jupyter, SPSS, Databricks, Excel (Advanced)
- **AI & LLM Engineering:** Claude, Claude Code, Gemini, GitHub Copilot, Prompt Engineering, AI Agents
- **Cloud:** AWS (ECS, EKS, Lambda, S3), Google BigQuery, OpenStack, OpenShift (Red Hat / enterprise K8s)
- **DevOps & Containers:** Docker, Podman, Kubernetes, Helm, Kind, k3s, Jenkins, Ansible, Maven, GitHub Actions, JMeter, SoapUI, Postman
- **Databases:** Oracle 19c, PostgreSQL, MySQL, Vertica, SQL Server, TimesTen
- **Observability:** Prometheus, Grafana, Elasticsearch, Kibana, Nagios
- **Architecture:** TOGAF, Solution Architecture, Data Architecture, Microservices, SOA, ESB, API Gateway, REST, SOAP, TM Forum SID
- **Integration:** IBM webMethods, TIBCO EAI/BPM, Siebel CRM, Salesforce, Oracle CRM
- **Methodologies:** SAFe 6, Scrum, Agile, DevOps, DevSecOps

Key Projects

- **Profit Erosion from E-Commerce Returns** (*MDA Capstone, 2026*) — Random Forest (AUC 0.9798), K-Means (Gini 0.409), log-linear regression (R^2 0.72) on Google BigQuery. Live Streamlit dashboard at profiterosion.streamlit.app; 512+ pytest tests and GitHub Actions CI/CD pipeline. Tools: Python, scikit-learn, BigQuery, Streamlit.
- **Warzone Armory** (*2025*) — AI-native full-stack web app built end-to-end with Claude Code: daily Python scraper via GitHub Actions cron, 256+ weapon database, playstyle filtering, analytics dashboard. Deployed on GitHub Pages. Tools: Python, GitHub Actions, Claude Code, Vanilla JS.
- **LR Presets Catalog** (*2025, in active development*) — Lightroom preset catalog with custom trilingual i18n (ES/EN/FR) built from scratch, Python/Pillow/OpenCV image processing pipeline, Chart.js tone-curve analytics, and Canvas 2D histogram comparison. Tools: Python, OpenCV, Chart.js, Claude Code, GitHub Pages.

Professional Experience

Solution Architect – Cloud-Native Integration & AI Systems · Corus Consulting

September 2025 – Present | Remote (Colombia / Latin America)

Architecting enterprise integration and AI systems for insurance and financial services clients across Colombia, Brazil, and Central America.

- Deployed IBM webMethods Integration Server on Kubernetes via custom Helm Charts — HA multi-pod layers, load-balanced ingress, and persistent volume configuration for production-ready containerised middleware.
- Built LLM-powered L2/L3 diagnostic agent (Copilot + Claude Code) covering 19 integration surfaces with SQL guardrails on Oracle 19c and structured skill playbooks — reduced ticket resolution time by 50%+.
- Engineered CI/CD pipelines (Jenkins, Ansible, Maven, GitHub Actions) and containerised sandbox environments (Docker, Podman, Kind, k3s, WSL2) enabling zero-drift deployment across dev, QA, and production.

Solution Architect – BSS/OSS Platform Delivery · Hewlett Packard Enterprise (HPE)

May 2010 – October 2024 | Colombia — Bell Canada, AT&T, Claro, Rogers Canada, Movistar, Tigo, ETB

Owned full-lifecycle delivery of BSS/OSS platform solutions for major carriers across the Americas on RHEL, with observability stacks (Prometheus, Grafana, Elasticsearch, Kibana) integrated across all deployments.

- Architected Hadoop/Spark/Hive ETL pipelines cutting data availability latency from 48+ hours to under 2 hours for Bell Canada and AT&T analytics environments.
- Reduced marketing campaign execution time from 10 days to 2 minutes by deploying Vertica columnar clustering with real-time MicroStrategy location intelligence for hyper-personalised targeting.
- Built Jenkins/Ansible/Maven CI/CD pipelines cutting deployment cycles from 10 days to 1 day across BSS/OSS platform releases for 4 carrier clients.
- Orchestrated OpenShift microservice deployments and OpenStack VM provisioning for Bell Canada, AT&T, and Claro — reducing infrastructure provisioning from 3 months to 2 days.

Earlier Career (2002–2010)

- **Integration Architect, Indra** (Mar 2008–May 2010) — TIBCO ESB for ETB Colombia: customer provisioning reduced 87% (15 days → 2 days), TM Forum SID-compliant.
- **Mediation Analyst, Millicom (Tigo)** (May 2005–Mar 2008) — Consolidated 3 legacy BSS mediation platforms (zero downtime, 3M+ subscribers). Java, Perl, C++ satellite applications.
- **Technical Consultant, Hewlett-Packard** (Jun 2004–May 2005) — Java J2EE payment reconciliation platform; reduced processing cycle from weeks to hours.
- **Software Developer, Proyectos Compartidos** (Dec 2002–Jun 2004) — Banking applications + neural network models for ATM cash forecasting (Davivienda, Bancolombia, Caja Social).

Education

- **Master of Data Analytics**, University of Niagara Falls Canada (2024–2026, completed)
President's Distinction List: Fall 2024, Winter 2025, Spring 2025, Fall 2025 · GPA 4.13 · \$1,000 academic scholarship each term
- **Bachelor of Electronic Engineering**, Universidad Nacional de Colombia (1996–2002)
GPA 3.7/5.0 · WES evaluated: equivalent to 4 years post-secondary education, Canada (April 2026)
- **Postgraduate Diploma in Project Management**, Universidad Santo Tomás, Colombia (2009–2010)
- **Postgraduate Diploma in Mobile Telecommunications**, Universidad Distrital Francisco José de Caldas, Colombia (2003–2004)

Certifications

- Certified SAFe® 6 Product Owner/Product Manager — Scaled Agile (2023)
- webMethods API Management Technical Sales Intermediate — IBM (Oct 2025)
- webMethods Hybrid Integration Sales Foundation — IBM (Nov 2025)
- webMethods Hybrid Integration Intermediate — IBM (Nov 2025)
- Microsoft Azure AI Essentials Professional Certificate — Microsoft & LinkedIn (in progress, 2026)
- Professional Engineering Licence — National Council of Electrical & Mechanical Engineering, Colombia (2010–Present)

Languages

Spanish (Native) · English (Professional)

Affiliations

Colombian National Professional Council of Electrical, Mechanical Engineering and Related Professions (2010–Present)

References available upon request.